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*
*      STAAD.Pro
*      Version 2007   Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   JAN 18, 2013
*      Time=   10:57:46
*
*      USER ID:
*****
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1. STAAD SPACE DXF IMPORT OF DRAWING1.DXF
INPUT FILE: 20121120 Pipe Bridge R1.STD
2. START JOB INFORMATION
3. ENGINEER DATE 09-MAY-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 -35.9989 0 -100.225; 4 -15.9989 0 -100.225; 5 -35.9989 0 -87.2249
9. 8 -15.9989 0 -87.2249; 9 -35.9989 5 -100.225; 10 -29.4989 5 -100.225
10. 11 -22.4989 5 -100.225; 12 -15.9989 5 -100.225; 13 -35.9989 5 -87.2249
11. 14 -29.4989 5 -87.2249; 15 -22.4989 5 -87.2249; 16 -15.9989 5 -87.2249
12. 17 -32.9989 5 -87.2249; 18 -35.9989 2 -87.2249; 19 -32.9989 5 -100.225
13. 20 -35.9989 2 -100.225; 21 -18.9989 5 -100.225; 22 -15.9989 2 -100.225
14. 23 -18.9989 5 -87.2249; 24 -15.9989 2 -87.2249; 26 -35.9989 5 -92.8149
15. 27 -15.9989 5 -92.8149; 28 -29.4989 5 -92.8149; 29 -22.4989 5 -92.8149
16. 30 -35.9989 0 -92.8149; 31 -15.9989 0 -92.8149; 32 -32.9989 5 -92.8149
17. 33 -35.9989 2 -92.8149; 34 -18.9989 5 -92.8149; 35 -15.9989 2 -92.8149
18. MEMBER INCIDENCES
19. 9 9 19; 10 10 11; 11 11 21; 12 9 26; 13 13 17; 14 14 15; 15 15 23; 16 12 27
20. 17 1 20; 18 5 18; 19 4 22; 20 8 24; 21 10 28; 22 11 29; 33 17 14; 34 18 13
21. 35 19 10; 36 20 9; 37 21 12; 38 22 12; 39 23 16; 40 24 16; 41 19 20; 42 17 18
22. 43 23 24; 44 21 22; 46 26 13; 47 27 16; 48 28 14; 49 29 15; 50 30 33; 52 31 35
23. 53 26 32; 54 28 29; 55 29 34; 58 10 29; 59 28 11; 60 28 15; 61 14 29; 62 11 27
24. 63 29 12; 64 29 16; 65 15 27; 68 26 5; 69 30 13; 70 16 31; 71 8 27; 72 27 4
25. 73 31 12; 74 32 28; 75 33 26; 76 34 27; 77 35 27; 78 32 33; 79 34 35
26. DEFINE MATERIAL START
27. ISOTROPIC STEEL
28. E 2.05E+008
29. POISSON 0.3
30. DENSITY 76.8195
31. ALPHA 1.2E-005
32. DAMP 0.03
33. END DEFINE MATERIAL
34. MEMBER PROPERTY JAPANESE
35. 17 TO 20 34 36 38 40 TABLE ST H428X407X20
36. 9 TO 11 13 TO 15 33 35 37 39 TABLE ST H428X407X20
37. 58 TO 65 68 TO 73 TABLE ST C200X90X8
38. 12 16 21 22 46 TO 49 TABLE ST H428X407X20
39. 50 52 75 77 TABLE ST H458X417X30
40. 53 TO 55 74 76 TABLE ST H498X432X45
41. 41 TO 44 78 79 TABLE ST H344X348X10
42. CONSTANTS
43. BETA 180 MEMB 68 71 73
44. BETA 90 MEMB 58 60 62 64
45. BETA 270 MEMB 59 61 63 65
46. MATERIAL STEEL ALL
47. SUPPORTS
48. 5 8 30 31 PINNED
49. 1 4 FIXED BUT KMX 0.001 KMY 0.001 KMZ 0.001
50. MEMBER RELEASE
51. 41 TO 44 78 79 START MY MZ
52. 41 TO 44 78 79 END MY MZ
53. 21 22 48 49 START MY MZ
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54. 21 22 48 49 END MY MZ
55. 12 16 46 47 START MY MZ
56. 12 16 46 47 END MY MZ
57. MEMBER TENSION
58. 58 TO 65 68 TO 73
59. ***** DEAD LOAD *****
60. LOAD 1 LOADTYPE NONE TITLE DL
61. SELFWEIGHT Y -1
62. MEMBER LOAD
63. 12 16 46 47 UNI GY -73.3
64. 21 22 49 UNI GY -62.39
65. 48 UNI GY -65.21
66. ***** FRICTIONAL LOAD *****
67. LOAD 2 LOADTYPE NONE TITLE FL
68. MEMBER LOAD
69. 12 16 46 47 UNI GX -7.33
70. 21 22 49 UNI GX -6.24
71. 48 UNI GX -6.521
72. ***** WIND LOAD X-DIRECTION 1*****
73. LOAD 3 LOADTYPE NONE TITLE WLX1
74. MEMBER LOAD
75. 58 TO 65 UNI GX -0.0765
76. 12 16 21 22 46 TO 49 UNI GX -0.364
77. 17 TO 20 34 36 38 40 UNI GX -0.346
78. 41 TO 44 78 79 UNI GX -0.2958
79. 68 TO 73 UNI GX -0.17
80. 50 52 75 77 UNI GX -0.355
81. ***** WIND LOAD X-DIRECTION 2*****
82. LOAD 4 LOADTYPE NONE TITLE WLX2
83. MEMBER LOAD
84. 58 TO 65 UNI GX 0.0765
85. 12 16 21 22 46 TO 49 UNI GX 0.364
86. 17 TO 20 34 36 38 40 UNI GX 0.346
87. 41 TO 44 78 79 UNI GX 0.2958
88. 68 TO 73 UNI GX 0.17
89. 50 52 75 77 UNI GX 0.355
90. ***** WIND LOAD Z-DIRECTION 1*****
91. LOAD 5 LOADTYPE NONE TITLE WLZ1
92. MEMBER LOAD
93. 58 TO 65 UNI GZ -0.0765
94. 9 TO 11 13 TO 15 33 35 37 39 UNI GZ -0.364
95. 53 TO 55 74 76 UNI GZ -0.4233
96. 17 TO 20 34 36 38 40 UNI GZ -0.364
97. 41 TO 44 78 79 UNI GZ -0.2924
98. 68 TO 73 UNI GZ -0.0765
99. 50 52 75 77 UNI GZ -0.3893
100. ***** WIND LOAD Z-DIRECTION 2*****
101. LOAD 6 LOADTYPE NONE TITLE WLZ2
102. MEMBER LOAD
103. 58 TO 65 UNI GZ 0.0765
104. 9 TO 11 13 TO 15 33 35 37 39 UNI GZ 0.364
105. 53 TO 55 74 76 UNI GZ 0.4233
106. 17 TO 20 34 36 38 40 UNI GZ 0.364
107. 41 TO 44 78 79 UNI GZ 0.2924
108. 68 TO 73 UNI GZ 0.0765
109. 50 52 75 77 UNI GZ 0.3893
110. ***** NOTIONAL LOAD X-DIRECTION 1*****
111. LOAD 7 LOADTYPE NONE TITLE NLX1
112. SELFWEIGHT X -0.015
113. MEMBER LOAD
114. 12 16 46 47 UNI GX -1.1
115. 21 22 49 UNI GX -0.94
116. 48 UNI GX -0.98
117. ***** NOTIONAL LOAD X-DIRECTION 2*****
118. LOAD 8 LOADTYPE NONE TITLE NLX2
119. SELFWEIGHT X 0.015
120. MEMBER LOAD
121. 12 16 46 47 UNI GX 1.1
122. 21 22 49 UNI GX 0.94
123. 48 UNI GX 0.98
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124. *****NOTIONAL LOAD Z-DIRECTION 1*****
125. LOAD 9 LOADTYPE NONE TITLE NLZ1
126. SELFWEIGHT Z -0.015
127. MEMBER LOAD
128. 12 16 46 47 UNI GZ -1.1
129. 21 22 49 UNI GZ -0.94
130. 48 UNI GZ -0.98
131. *****NOTIONAL LOAD Z-DIRECTION 2*****
132. LOAD 10 LOADTYPE NONE TITLE NLZ2
133. SELFWEIGHT Z 0.015
134. MEMBER LOAD
135. 12 16 46 47 UNI GZ 1.1
136. 21 22 49 UNI GZ 0.94
137. 48 UNI GZ 0.98
138. ***** LOAD COMBINATION - SERVICE *****
139. LOAD COMB 11 1.0DL+1.0FL
140. 1 1.0 2 1.0
141. LOAD COMB 12 1.0DL+1.0FL+1.0WLX1
142. 1 1.0 2 1.0 3 1.0
143. LOAD COMB 13 1.0DL+1.0FL+1.0WLX2
144. 1 1.0 2 1.0 4 1.0
145. LOAD COMB 14 1.0DL+1.0FL+1.0WLZ1
146. 1 1.0 2 1.0 5 1.0
147. LOAD COMB 15 1.0DL+1.0FL+1.0WLZ2
148. 1 1.0 2 1.0 6 1.0
149. LOAD COMB 16 1.0DL+1.0FL+1.0NLX1
150. 1 1.0 2 1.0 7 1.0
151. LOAD COMB 17 1.0DL+1.0FL+1.0NLX2
152. 1 1.0 2 1.0 8 1.0
153. LOAD COMB 18 1.0DL+1.0FL+1.0NLZ1
154. 1 1.0 2 1.0 9 1.0
155. LOAD COMB 19 1.0DL+1.0FL+1.0NLZ2
156. 1 1.0 2 1.0 10 1.0
157. ***** LOAD COMBINATION - ULTIMATE *****
158. LOAD COMB 20 1.0DL+1.4FL
159. 1 1.0 2 1.4
160. LOAD COMB 21 1.4DL+1.4FL
161. 1 1.4 2 1.4
162. LOAD COMB 22 1.0DL+1.4FL+1.4WLX1
163. 1 1.0 2 1.4 3 1.4
164. LOAD COMB 23 1.0DL+1.4FL+1.4WLX2
165. 1 1.0 2 1.4 4 1.4
166. LOAD COMB 24 1.0DL+1.4FL+1.4WLZ1
167. 1 1.0 2 1.4 5 1.4
168. LOAD COMB 25 1.0DL+1.4FL+1.4WLZ2
169. 1 1.0 2 1.4 6 1.4
170. LOAD COMB 26 1.0DL+1.4FL+1.4NLX1
171. 1 1.0 2 1.4 7 1.4
172. LOAD COMB 27 1.0DL+1.4FL+1.4NLX2
173. 1 1.0 2 1.4 8 1.4
174. LOAD COMB 28 1.0DL+1.4FL+1.4NLZ1
175. 1 1.0 2 1.4 9 1.4
176. LOAD COMB 29 1.0DL+1.4FL+1.4NLZ2
177. 1 1.0 2 1.4 10 1.4
178. *****ANALYSIS*****
179. PERFORM ANALYSIS

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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 30/ 55/ 6

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 162

**START ITERATION NO. 2
**START ITERATION NO. 3

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**START ITERATION NO.            4
**START ITERATION NO.            5
**START ITERATION NO.            6

**NOTE-Tension/Compression converged after    6 iterations, Case=    1

**START ITERATION NO.            2
**START ITERATION NO.            3

**NOTE-Tension/Compression converged after    3 iterations, Case=    2

**START ITERATION NO.            2
**START ITERATION NO.            3

**NOTE-Tension/Compression converged after    3 iterations, Case=    3

**START ITERATION NO.            2
**START ITERATION NO.            3
**START ITERATION NO.            4

**NOTE-Tension/Compression converged after    4 iterations, Case=    4

**START ITERATION NO.            2
**START ITERATION NO.            3

**NOTE-Tension/Compression converged after    3 iterations, Case=    5

**START ITERATION NO.            2

**NOTE-Tension/Compression converged after    2 iterations, Case=    6

**START ITERATION NO.            2
**START ITERATION NO.            3

**NOTE-Tension/Compression converged after    3 iterations, Case=    7

**START ITERATION NO.            2

**NOTE-Tension/Compression converged after    2 iterations, Case=    8

**START ITERATION NO.            2
**START ITERATION NO.            3

**NOTE-Tension/Compression converged after    3 iterations, Case=    9

**START ITERATION NO.            2
**START ITERATION NO.            3

**NOTE-Tension/Compression converged after    3 iterations, Case=    10

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